

Project Executable Files

Field	Details
Starting Date	08 October 2026
Team ID	UNM1611Firefox
Project Name	AI STUDY BUDDY API

Step 1: Submission Checklist

Checklist Item	Project evidence	Status	Marks / Evaluation
Source code included	Node.js/Express API source, routes, controllers, models, middleware and utilities	Yes	Evaluator entry
Configuration included	.env is present in ZIP; secrets should not be shared in public submission	Yes / review secrets	Evaluator entry
Dependency file	package.json and package-lock.json	Yes	Evaluator entry
Database integration	Mongoose/MongoDB utility and models	Yes	Evaluator entry
AI integration	Google Generative AI utility using Gemini	Yes	Evaluator entry
Run instructions	Documented below	Yes	Evaluator entry

Step 2: File / Folder Structure

Path	Purpose
Code Files/index.js	Application entry point
Code Files/package.json	Project metadata, scripts and dependencies
Code Files/src/routes/auth.js	Authentication routes
Code Files/src/routes/materials.js	Study-material and AI feature routes
Code Files/src/routes/admin.js	Admin routes
Code Files/src/controllers/authController.js	Register/login/refresh/logout logic
Code Files/src/controllers/materialController.js	Upload, retrieval, deletion and AI operations
Code Files/src/controllers/adminController.js	Admin user/material statistics and management
Code Files/src/models/User.js	User schema
Code Files/src/models/Material.js	Study-material schema
Code Files/src/middleware/auth.js	JWT authentication and role checks
Code Files/src/middleware/upload.js	Multer upload configuration and validation
Code Files/src/utlis/gemini.js	Gemini API integration
Code Files/src/utlis/tokens.js	JWT access/refresh token handling
Code Files/src/utlis/db.js	MongoDB connection utility

Step 3: Deployment / Access Details

Application type: Node.js + Express REST API. Database: MongoDB through Mongoose. AI provider: Google Gemini via @google/generative-ai. Authentication: JWT access and refresh tokens. File uploads: Multer.

Environment variables required by the source include the database connection configuration, JWT secrets/configuration and Gemini API configuration. The exact values must be supplied locally and must not be embedded in the submitted documentation.

The supplied ZIP contains a .env file. Do not publish real credentials from that file to GitHub or include them in screenshots.

Step 4: Run Instructions

Step	Instruction
1	Install a supported Node.js runtime and MongoDB access.
2	Open the Code Files directory.
3	Install dependencies with npm install.
4	Configure the required environment variables in .env.
5	Start the application using the project's configured start command / entry point.
6	Use an API client such as Postman/Insomnia to call the authentication, material, AI and admin endpoints.

Important source-code note: package.json defines the start script as "nodemon server.js", while the supplied project listing contains index.js as the entry point and no server.js file. This is a packaging/run-configuration inconsistency that should be corrected before claiming a clean one-command execution.

Step 5: Known Issues / Limitations

Issue / Limitation	Impact	Recommended action
No automated test suite; test script is placeholder	Execution evidence and regression coverage are limited	Add unit/integration/API tests
Start script references server.js while supplied source has index.js	npm start may fail as packaged	Align package.json with actual entry file
PDF extension is accepted by upload filter, but controller reads file as UTF-8 text	Binary PDF content may not be extracted correctly	Add a real PDF text extraction library or restrict supported formats
Logout does not revoke JWTs server-side	A previously issued token can remain usable until expiry	Use token revocation/rotation strategy if required
AI JSON is parsed directly	Malformed Gemini output can cause parsing failures	Validate generated JSON against a schema
No performance execution logs supplied	Cannot claim measured latency/throughput	Run load/performance tests and attach evidence